

Approximation and Taylor's Theorem

Question Bank

Engineering Mathematics–I (CSE-1001)

BTech 1st Year, IIIT Bhopal

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Instructions

This bank accompanies the note *Approximation and Taylor's Theorem*. Work through a part only after you have read the matching sections of the note.

How the parts are meant to be used:

1. **Part A (MCQ)** — rapid revision. One minute per question.
2. **Part B (True or False)** — every answer must carry a reason or a counterexample.
3. **Part C (Short answer)** — the routine computations that make up the bulk of an examination paper.
4. **Part D (Long answer)** — statements, proofs and multi-step problems, written out as you would in the examination.
5. **Part E (Critical thinking)** — traps, borderline cases and conceptual questions.

A note on calculators. Several questions ask you to compare an approximation with the true value. Use a calculator for the true value only — the approximation itself must be computed by hand, since that is what is being examined. And in every error question, say clearly whether you are giving a *bound* (guaranteed) or an *actual error* (computed against the true value). They are different things and carry different marks.

Answers and full worked solutions are issued as a separate document, *Approximation and Taylor's Theorem — Answer Key*.

Part A: Multiple Choice Questions

44 questions $\times$ 1 mark

Choose the one correct option.

Q1. The linear approximation to f at the point a is

- (a) $f(a)$ (b) $f'(a)(x - a)$ (c) $f(a) + f'(a)(x - a)$ (d) $f(a) + f'(x)(x - a)$

Q2. Geometrically, the linear approximation to f at a is

- (a) a secant line through two points of the graph
(b) the tangent line to the graph at $(a, f(a))$
(c) the normal line at $(a, f(a))$
(d) a chord of the graph

Q3. The polynomial P_1 is characterised by

- (a) $P_1(a) = f(a)$ only
(b) $P_1(a) = f(a)$ and $P_1'(a) = f'(a)$
(c) $P_1''(a) = f''(a)$
(d) $P_1(x) = f(x)$ for all x

Q4. Near $x = 0$, the linear approximation of $\sin x$ is

- (a) x (b) 1 (c) $1 - \frac{x^2}{2}$ (d) $x - \frac{x^3}{6}$

Q5. Near $x = 0$, the linear approximation of $\cos x$ is

- (a) 1 (b) x (c) 0 (d) $1 - \frac{x^2}{2}$

Q6. Near $x = 0$, the linear approximation of e^x is

- (a) x (b) 1 (c) $1 + x + \frac{x^2}{2}$ (d) $1 + x$

Q7. Near $x = 0$, the linear approximation of $\ln(1 + x)$ is

- (a) x (b) $1 + x$ (c) $x - \frac{x^2}{2}$ (d) $\ln x$

Q8. Near $x = 0$, the linear approximation of $(1 + x)^r$ is

- (a) $1 + x$ (b) $r + x$ (c) $1 + rx$ (d) $1 + rx + \frac{r(r-1)}{2}x^2$

Q9. Using $a = 4$, the linear approximation gives $\sqrt{4.05} \approx$

- (a) 2.0125 (b) 2.05 (c) 2.0025 (d) 2.01

Q10. The base point a should be chosen so that

- (a) $a = 0$, always
(b) $f(a)$ and $f'(a)$ are known exactly, and a is close to x
(c) $f''(a) = 0$
(d) $a = x$

Q11. Near $x = 0$, $\sqrt{1+x} \approx$

- (a) $1 + \frac{x}{2}$ (b) $1 + x$ (c) $1 - \frac{x}{2}$ (d) $1 + 2x$

Q12. The quadratic approximation to f at a is

- (a) $f(a) + f'(a)(x - a)$
(b) $f(a) + f'(a)(x - a) + f''(a)(x - a)^2$
(c) $f(a) + f'(a)(x - a) + \frac{f''(a)}{2}(x - a)^2$
(d) $f(a) + \frac{f''(a)}{2}(x - a)^2$

Q13. The factor $\frac{1}{2}$ in the quadratic term is there because

- (a) it makes the formula exact for quadratic functions
(b) of tradition
(c) it halves the error
(d) of the chain rule

Q14. Near $x = 0$, the quadratic approximation of $\cos x$ is

- (a) 1 (b) $1 + \frac{x^2}{2}$ (c) $1 - \frac{x^2}{2}$ (d) x

Q15. Near $x = 0$, the quadratic approximation of e^x is

- (a) $1 + x$ (b) $1 + x + x^2$ (c) $1 + x + \frac{x^2}{2}$ (d) $1 + \frac{x^2}{2}$

Q16. Near $x = 0$, the quadratic approximation of $\ln(1+x)$ is

- (a) x (b) $x + \frac{x^2}{2}$ (c) $x - x^2$ (d) $x - \frac{x^2}{2}$

Q17. Near $x = 0$, the quadratic approximation of $(1+x)^r$ is

- (a) $1 + rx$
(b) $1 + rx + \frac{r(r-1)}{2}x^2$
(c) $1 + rx + \frac{r}{2}x^2$
(d) $1 + rx + \frac{r^2}{2}x^2$

Q18. The quadratic approximation of $\sin x$ at $a = 0$ is

- (a) x (b) $x - \frac{x^3}{6}$ (c) $1 - \frac{x^2}{2}$ (d) $x + \frac{x^2}{2}$

Q19. When two linear approximations are multiplied together, the resulting x^2 term should be

- (a) kept (b) doubled (c) discarded (d) halved

Q20. Near $x = 0$, the linear approximation of $\frac{e^{-2x}}{\sqrt{1+x}}$ is

- (a) $1 - \frac{5}{2}x$ (b) $1 - 2x$ (c) $1 - \frac{3}{2}x$ (d) $1 + \frac{5}{2}x$

Q21. The error in the linear approximation satisfies

$$(a) |E_1| \leq M(x-a) \quad (b) |E_1| \leq \frac{M}{6}|x-a|^3 \quad (c) |E_1| \leq \frac{M}{2}(x-a)^2 \quad (d) |E_1| \leq M$$

Q22. In that bound, M must be

- (a) the value $f''(a)$
- (b) an upper bound for $|f''|$ on the whole interval between a and x
- (c) the value $f'(a)$
- (d) the maximum of f

Q23. The error in the quadratic approximation satisfies

$$(a) |E_2| \leq \frac{M}{2}(x-a)^2 \quad (b) |E_2| \leq \frac{M}{24}|x-a|^4 \quad (c) |E_2| \leq M|x-a|^3 \quad (d) |E_2| \leq \frac{M}{6}|x-a|^3$$

Q24. Halving the distance $|x-a|$ divides the bound on the linear approximation error by

- (a) 4
- (b) 2
- (c) 8
- (d) 16

Q25. The Extended Mean Value Theorem is Taylor's Theorem in the case

- (a) $n = 0$
- (b) $n = 1$
- (c) $n = 2$
- (d) $n \rightarrow \infty$

Q26. Taylor's Theorem with $n = 0$ reduces to

- (a) Rolle's theorem
- (b) the Intermediate Value Theorem
- (c) L'Hôpital's rule
- (d) the Mean Value Theorem

Q27. The Taylor polynomial of degree n about a is

$$(a) \sum_{k=0}^n f^{(k)}(a)(x-a)^k \quad (b) \sum_{k=0}^n \frac{f^{(k)}(a)}{k!}(x-a)^k \quad (c) \sum_{k=0}^n \frac{f^{(k)}(x)}{k!}(x-a)^k \quad (d) \sum_{k=0}^n \frac{f(a)}{k!}x^k$$

Q28. The Lagrange form of the remainder is

$$(a) \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \quad (b) \frac{f^{(n)}(c)}{n!}(x-a)^n \quad (c) \frac{f^{(n+1)}(a)}{(n+1)!}(x-a)^{n+1} \quad (d) 0$$

Q29. In Taylor's Theorem, the point c

- (a) equals a
- (b) equals x
- (c) lies strictly between a and x , and is generally unknown
- (d) may be chosen arbitrarily

Q30. The standard proof of Taylor's Theorem uses

- (a) the Intermediate Value Theorem
- (b) Rolle's theorem
- (c) L'Hôpital's rule
- (d) the Mean Value Theorem for integrals

Q31. If $|f^{(n+1)}(t)| \leq M$ between a and x , then

$$\frac{(a) \quad |R_n|}{M|x-a|^n} \leq (b) \quad |R_n| \leq M \quad (c) \quad \frac{|R_n|}{M|x-a|^{n+1}} \leq (d) \quad |R_n| \leq \frac{M}{(n+1)!}$$

Q32. Using $P_2(x) = 1 - \frac{x^2}{2}$ for $\cos x$ at $x = 0.6$, the error bound from the Remainder Estimation Theorem is

- (a) 0.036
- (b) 0.36
- (c) 0.0036
- (d) 0.006

Q33. $\sin x$ may be replaced by $x - \frac{x^3}{6}$ with error at most 4×10^{-3} for approximately

- (a) $|x| \leq 0.056$
- (b) $|x| \leq 0.556$
- (c) $|x| \leq 5.56$
- (d) all x

Q34. The Taylor series of f converges to $f(x)$ if and only if

- (a) $R_n(x) \rightarrow 0$ as $n \rightarrow \infty$
- (b) f is continuous
- (c) $P_n(x) \rightarrow 0$
- (d) $f^{(n)}(a) \rightarrow 0$

Q35. A Maclaurin series is a Taylor series with

- (a) $a = 1$
- (b) $a = 0$
- (c) $a = x$
- (d) $a = \infty$

Q36. The Maclaurin series of e^x converges for

- (a) $|x| < 1$
- (b) $x > 0$
- (c) every real x
- (d) $|x| \leq 1$

Q37. $\sum_{n=0}^{\infty} \frac{1}{n!}$ equals

- (a) e
- (b) $e - 1$
- (c) 1
- (d) 2

Q38. The Maclaurin series of $\frac{1}{1-x}$ converges for

- (a) every real x
- (b) $|x| < 1$
- (c) $|x| \leq 1$
- (d) $x > 0$

Q39. For an alternating Taylor series, the truncation error is at most

- (a) the last term kept
- (b) zero
- (c) the sum of all remaining terms
- (d) the first term omitted

- Q40.** The inequality $1 - \frac{x^2}{2} \leq \cos x$ for all real x is most directly proved by
- (a) the Mean Value Theorem
 - (b) Taylor's Theorem with $n = 2$, arguing on the sign of the remainder
 - (c) the Intermediate Value Theorem
 - (d) L'Hôpital's rule
- Q41.** If $f'(a) = f''(a) = f'''(a) = 0$ and $f^{(4)}(a) > 0$, then f has at a
- (a) a local maximum
 - (b) a point of inflection
 - (c) a local minimum
 - (d) no extremum
- Q42.** If $f'' \geq 0$ on an interval, the graph of f lies
- (a) on or above each of its tangent lines
 - (b) below each of its tangent lines
 - (c) exactly on its tangent lines
 - (d) sometimes above, sometimes below
- Q43.** For a function with $f'' \geq 0$, the linear approximation is
- (a) an overestimate
 - (b) an underestimate
 - (c) exact
 - (d) undetermined
- Q44.** The first non-zero correction term to the approximation $\sin x \approx x$ is of order
- (a) x^2
 - (b) x
 - (c) x^3
 - (d) x^4

Part B: True or False — with justification

10 questions $\times$ 1 mark

State whether each is true or false. **A verdict without a reason or a counterexample earns no marks.**

- Q1.** The linear approximation is exact when f is itself a linear function.
- Q2.** The quadratic approximation is exact when f is a quadratic polynomial.
- Q3.** If f has derivatives of all orders at 0, then its Maclaurin series converges to $f(x)$ for every real x .
- Q4.** In Taylor's Theorem, the point c can in general be computed explicitly.
- Q5.** In the bound $|E_1| \leq \frac{M}{2}(x-a)^2$, one may take $M = |f''(a)|$.
- Q6.** Taylor's Theorem with $n = 0$ is precisely the Mean Value Theorem.
- Q7.** If $f''(a) = 0$, then a is a point of inflection of f .
- Q8.** The Maclaurin series of $\cos x$ contains only even powers of x .
- Q9.** Doubling the degree of a Taylor polynomial always at least halves the error.

Q10. For a convex function, the tangent line at any point underestimates the function.

Part C: Short Answer Questions

20 questions $\times$ 3 marks

- Q1.** Define the linear approximation of f at a , and find it for $f(x) = \sqrt{x}$ at $a = 9$.
- Q2.** Using the previous result, compute $\sqrt{9.1}$ and compare with the true value $3.016621\dots$
- Q3.** Compute $\sin(0.02)$ and $e^{0.01}$ by linear approximation.
- Q4.** Prove that $(1+x)^r \approx 1+rx$ near $x=0$, and use it to estimate $\sqrt[4]{1.1}$.
- Q5.** Find the linear approximation of $\ln x$ near $x=1$, and deduce $\ln(1+u) \approx u$ by changing the base point.
- Q6.** Find the linear approximation of $\frac{e^{-2x}}{\sqrt{1+x}}$ near $x=0$ by combining standard approximations, and hence write down $f(0)$ and $f'(0)$.
- Q7.** Define the quadratic approximation of f at a , and explain why the coefficient $\frac{1}{2}$ appears.
- Q8.** Compute $\ln(1.1)$ and $e^{0.1}$ by quadratic approximation.
- Q9.** Find the quadratic approximation of $\cos(0.2)$ and compare with the true value.
- Q10.** Compute $\sqrt{4.05}$ by quadratic approximation, and state how much better it is than the linear estimate 2.0125 .
- Q11.** Find the quadratic approximation of $\frac{e^{-2x}}{\sqrt{1+x}}$ near $x=0$.
- Q12.** State the error bound for the linear approximation, and use it to bound the error in $\sqrt{1.1} \approx 1.05$, taking $a=1$.
- Q13.** State the error bound for the quadratic approximation, and use it to bound the error in the quadratic estimate of $\sqrt{1.1}$.
- Q14.** Write down the Taylor polynomial P_3 for
- (i) $f(x) = e^x$ about $a=0$;
 - (ii) $f(x) = \frac{1}{x}$ about $a=1$.
- Q15.** State Taylor's Theorem with the Lagrange form of the remainder, identifying clearly what is known and what is unknown about the point c .
- Q16.** State the Remainder Estimation Theorem and use it to estimate the error when $P_2(x) = 1 - \frac{x^2}{2}$ is used for $\cos x$ at $x=0.6$.
- Q17.** Estimate the maximum error when $f(x) = \sqrt{x}$ is approximated by its Taylor polynomial with $n=2$ about $a=4$, on $4 \leq x \leq 4.2$.

- Q18.** For approximately what values of x may $\sin x$ be replaced by $x - \frac{x^3}{6}$ with an error no greater than 4×10^{-3} ?
- Q19.** Write down the Maclaurin series of e^x , $\sin x$ and $\cos x$, and state for which x each is valid.
- Q20.** Find the quadratic approximation of $f(x) = \ln(\sin x)$ near $x = \frac{\pi}{2}$.

Part D: Long Answer Questions

12 questions $\times$ 6 marks

- Q1.** Define the linear approximation of a function at a point. Derive, with proof, the five standard linear approximations near $x = 0$: $\sin x$, $\cos x$, e^x , $\ln(1+x)$ and $(1+x)^r$.
- Q2.** Explain the method of finding an approximation by *combining* standard approximations rather than by differentiating. Apply it to $f(x) = \frac{e^{-2x}}{\sqrt{1+x}}$ to first order and then to second order, explaining at each stage which terms are discarded and why.
- Q3.** Define the quadratic approximation. Show that the coefficient $\frac{1}{2}$ is exactly what makes the formula reproduce a quadratic polynomial exactly. Hence derive the five standard quadratic approximations near $x = 0$.
- Q4.** State the Extended Mean Value Theorem. Deduce the error bounds

$$|E_1(x)| \leq \frac{M}{2}(x-a)^2 \quad \text{and} \quad |E_2(x)| \leq \frac{M}{6}|x-a|^3,$$

saying precisely what M means in each. Apply both to $\sqrt{1.1}$ with $a = 1$, and compare each bound with the actual error.

- Q5.** Compute $\sqrt{4.05}$ by the linear and by the quadratic approximation about $a = 4$. Tabulate both against the true value 2.01246118, give the error in each case, and comment on the improvement obtained from one extra term.
- Q6.** State Taylor's Theorem with the Lagrange remainder, and prove it.
(Hint: choose M so that the formula holds with remainder $\frac{M}{(n+1)!}(x-a)^{n+1}$, build the auxiliary function $g(t) = \sum_{k=0}^n \frac{f^{(k)}(t)}{k!}(x-t)^k + \frac{M}{(n+1)!}(x-t)^{n+1}$, and apply Rolle's theorem.)
- Q7.** Show that Taylor's Theorem contains both the Mean Value Theorem and the Extended Mean Value Theorem as special cases, identifying the value of n in each case and writing out the resulting statement.
- Q8.** State the Remainder Estimation Theorem and prove it from Taylor's Theorem. Use it to settle all three of:
- (i) the error in approximating $\cos(0.6)$ by $1 - \frac{x^2}{2}$;

- (ii) the maximum error for $\sqrt{x}$ with $n = 2$, $a = 4$, on $[4, 4.2]$;
- (iii) the range of x for which $\sin x \approx x - \frac{x^3}{6}$ has error at most 4×10^{-3} .

- Q9.** Prove that the Maclaurin series of e^x converges to e^x for every real x , showing carefully that $R_n(x) \rightarrow 0$.
- Q10.** Prove that the Maclaurin series of $\sin x$ converges to $\sin x$ for every real x . Write down also the Maclaurin series of $\cos x$ and of $\frac{1}{1-x}$, stating the range of validity of each.
- Q11.** Use Taylor's Theorem to prove that
- (i) $1 - \frac{x^2}{2} \leq \cos x$ for every real x ;
 - (ii) for every $k \in \mathbb{N}$ and every $x > 0$,

$$x - \frac{x^2}{2} + \cdots - \frac{x^{2k}}{2k} < \ln(1+x) < x - \frac{x^2}{2} + \cdots + \frac{x^{2k+1}}{2k+1}.$$

- Q12.** State the general n -th derivative test for a local maximum or minimum, and prove it using Taylor's Theorem. Apply it to $f(x) = x^4$ and $f(x) = x^3$ at $x = 0$. Prove also that if $f'' \geq 0$ on an interval then the graph of f lies on or above each of its tangent lines.

Part E: Critical Thinking Questions

8 questions $\times$ 5 marks

- Q1.** A student computes $\sqrt{4.05}$ by taking the base point $a = 0$, reasoning that “0 is the standard base point”. Explain in detail what goes wrong, state the two separate conditions a good base point must satisfy, and say which one fails here.
- Q2.** For $\cos x$ approximated by $1 - \frac{x^2}{2}$ at $x = 0.6$, the Remainder Estimation Theorem gives the bound 0.036, whereas the actual error is 0.0053 — nearly seven times smaller. Is the theorem wrong? Explain the difference between a *bound* and an *estimate*, and show that the alternating series estimate gives a sharper answer here.
- Q3.** The linear and the quadratic approximations of $\sin x$ at $a = 0$ are both equal to x . Explain why. What does this say about the accuracy of $\sin x \approx x$ compared with $\cos x \approx 1$, and how is that visible in the error bounds?
- Q4.** The function $f(x) = \frac{1}{1-x}$ has derivatives of all orders at every point except $x = 1$. Show that its Maclaurin series converges only for $|x| < 1$. At $x = 2$ the function is perfectly well behaved, with $f(2) = -1$; explain why the series nevertheless fails there, and what feature of f is responsible.
- Q5.** One student multiplies $e^x \approx 1 + x$ by $\sin x \approx x$, keeps the x^2 term and reports $e^x \sin x \approx x + x^2$. Another discards it and reports x . Both are working near $x = 0$. Decide who is right, and be precise about the question each is answering.

- Q6.** In Taylor's Theorem the point c depends on *both* x and n . A student writes:
“ $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}x^{n+1} \rightarrow 0$ as $n \rightarrow \infty$, because factorials beat powers.” For $f(x) = e^x$ the conclusion is correct, but the reasoning as written is incomplete. Identify what has been assumed without justification, and supply the missing step.
- Q7.** The Maclaurin series of $\sin x$ converges for every real x , including $x = 100$. Explain why no numerical library evaluates $\sin(100)$ by summing that series directly, and describe what is done instead. What general lesson does this carry about the difference between mathematical convergence and numerical usefulness?
- Q8.** The second-derivative test is silent at $x = 0$ for both $f(x) = x^4$ and $g(x) = x^3$, since $f''(0) = g''(0) = 0$. Use Taylor's Theorem to distinguish the two cases completely, and state the general rule that decides such questions.

Answers: see **Approximation and Taylor's Theorem — Answer Key**